Measuring Image Dependence, Not Just Response Sensitivity: Controlled Ablation Tests on Brain-MRI Question Answering

Anonymous ACL submission

Abstract

Accuracy changes after removing or swapping an image do not by themselves show that a multimodal LLM uses the image correctly; they may reflect response sensitivity alone. We test image dependence by comparing ablation against the answer prior, scoring donor-image swaps against the donor key, and requiring an image-only positive control on brain-MRI questions from the LUMIERE glioblastoma cohort.

A reference model reads sequence and lesion side from the image (95.7% and 100.0%, versus 21.7% and 36.8% without it) and tracks donor swaps, showing the tasks are feasible. Four open-weight models show limited image dependence: only one image-effect test survives multiplicity correction, its accuracy remains below the most common key's share, no donor-specific tracking survives multiplicity correction, and none exceeds the best category lookup on the management rule. Two bf16 reruns remain near chance on the control, making 4-bit serving an unlikely sole explanation.

A motivating audit finds that a slice-based enhancement rule matches the expert RANO rating for 38% of 197 determinate follow-ups, so those ratings are not treated as validated targets. We measure observable image-dependent behavior, not internal image use; keys are segmentation- or rule-derived and not clinician-validated.

1 Introduction

Medical multimodal LLMs are often evaluated by final-answer accuracy, which cannot distinguish image-based reasoning from reliance on textual cues, option patterns, or guessing. Such shortcuts are well documented: models exploit annotation artifacts in natural language inference (Gururangan et al., 2018), syntactic heuristics (McCoy et al., 2019), and unfaithful stated reasoning (Turpin et al., 2023). In chest-radiograph question answering, models can rely on linguistic patterns or irrelevant image regions (Nguyen et al., 2025). Image ablation or swapping is informative only when the displayed image contains enough evidence to recover the answer.

We ask when an image intervention measures image dependence rather than merely response sensitivity. An accuracy difference between an image arm and a text-only arm describes those two conditions, but a weak text-only arm leaves it insufficient as evidence that the image is read correctly, and an answer that changes when the image is swapped need not move toward the answer the new image supports. We therefore read the ablation together with three additions: reference values for each phase; a matched donor swap scored against the donor's key with the no-image prior subtracted; and an audit that compares the key against an independent expert rating, which checks agreement between two labeling procedures rather than confirming that the key is recoverable from the displayed slices. Protocols that perturb the text or the image and report the change in accuracy, such as HEAL-MedVQA (Nguyen et al., 2025), address the same concern; we did not run their protocol, so the comparison is by design, not measured, and Section 5 gives a worked case in which the ablation alone points to image use and the additions do not. We instantiate the questions on the five OmniBrain-Bench phases (Peng et al., 2026) as same-patient chains from the LUMIERE glioblastoma cohort (Suter et al., 2022): anatomical and imaging assessment (AIA), lesion identification and localization (LIL), diagnostic synthesis and causal reasoning (DSCR), prognostic judgment and risk forecasting (PJRF), and therapeutic cycle management (TCM); Figure 1 summarizes the design. The chain is an extension that lets us also test whether a model uses stated facts and upstream context in the management step. A motivating audit of the expert RANO ratings found that they cannot be treated as validated targets for the displayed slices (Section 3.1),

which motivates an evaluation with explicit key provenance and fixed controls (Sections 3.2–4).

The paper makes three contributions:

- A protocol that keeps three outcomes apart, accuracy benefit, output sensitivity and donor tracking, and reads image ablation against the answer prior, with a worked case (Section 5) in which the ablation alone credits Llama-4-Scout with image use on the diagnosis category although its accuracy with the image stays below the most common key's share and its donor gain is not separable from a random donor assignment.
- An item set with an explicit origin for each key, built on LUMIERE, and a motivating audit of the expert labels: an enhancement-based rule reproduces the expert RANO category for 38% of the 197 follow-ups on which it is determinate (coverage is reported separately: it is undetermined for 44 of 241), so we do not treat those ratings as validated targets for the displayed slices; the audit does not show which side is wrong. The released OmniBrainBench groupings do not give same-patient chains (only 2 verifiable cross-phase links among 6,823 questions).
- An evaluation of four open-weight models and one reference model under a prespecified control suite. The reference model shows image-dependent performance on sequence and lesion-side questions, whereas the open models are limited and inconsistent; on the diagnosis-category task the most common key covers 58.7% of items, so accuracy there is hard to separate from a class prior even for the reference model. As an extension, the chain tests the use of stated facts and upstream context on the management step.

2 Related Work

Shortcuts and faithfulness. Models can reach correct labels without performing the intended reasoning. Hypothesis-only baselines exploit annotation artifacts in NLI datasets (Poliak et al., 2018; Gururangan et al., 2018), models adopt heuristics that fail on HANS (McCoy et al., 2019), and biased input features can alter chain-of-thought answers without appearing in the stated rationale (Turpin et al., 2023).

Answering without the image. Visual question-answering models often exploit language priors. Goyal et al. (2017) countered this by pairing questions with images yielding different answers, while Chen et al. (2024) show that many multimodal benchmark items do not require visual content. Similar failures appear in medicine: Yan et al. (2025) report severe degradation under adversarial diagnostic counterparts, Jin et al. (2024) find flawed rationales in 35.5% of GPT-4V's correct answers, and Nguyen et al. (2025) (HEAL-MedVQA) use textual and visual perturbations to localize shortcuts in chest-radiograph QA. We extend this logic to brain-MRI questions and add a donor swap scored against the donor's key with the no-image prior subtracted, per-phase reference values, and a key-recoverability audit. Unlike correctness on complementary pairs (Goyal et al., 2017), which requires the model to answer each image correctly, this statistic measures movement toward what the swapped image supports and so is informative even for models that are rarely correct, as ours are; we did not run HEAL-MedVQA's protocol, so that comparison is one of design. The combination adds discrimination a raw gain lacks: a non-specific image shift and a noisy tracker both reach +19.0 points of gain and change a comparable share of answers, yet the conditional donor-label permutation flags them at 0% and 78% (Table 6). The same-patient chain extends the questions to interventions on supplied clinical facts. UniMod (Gu et al., 2026) addresses modality imbalance during training by supervising image-only, text-only, and fused predictions separately; we instead evaluate existing models for observable image dependence.

Brain benchmarks and longitudinal data. OmniBrainBench (Peng et al., 2026) contains 6,823 closed-ended brain-imaging questions spanning five clinical phases and 15 modalities. Its supplement describes a patient-level ADNI extension, but we did not evaluate it. LUMIERE (Suter et al., 2022) and MU-Glioma-Post (Mahmoud et al., 2025) provide longitudinal glioma imaging, treatment, and outcome data but no question layer.

3 Building Chains from LUMIERE

Why not OmniBrainBench. A chain needs one patient's phases. We checked whether any image is shared across questions of different phases within a grouped case: of 6,823 closed-ended questions only 2 verifiable cross-phase patient links exist, and a second method based on filename identifiers agreed; the counts are lower bounds (Appendix A). Each sub-corpus was built for one task, so a patient is essentially asked one phase-type of ques-

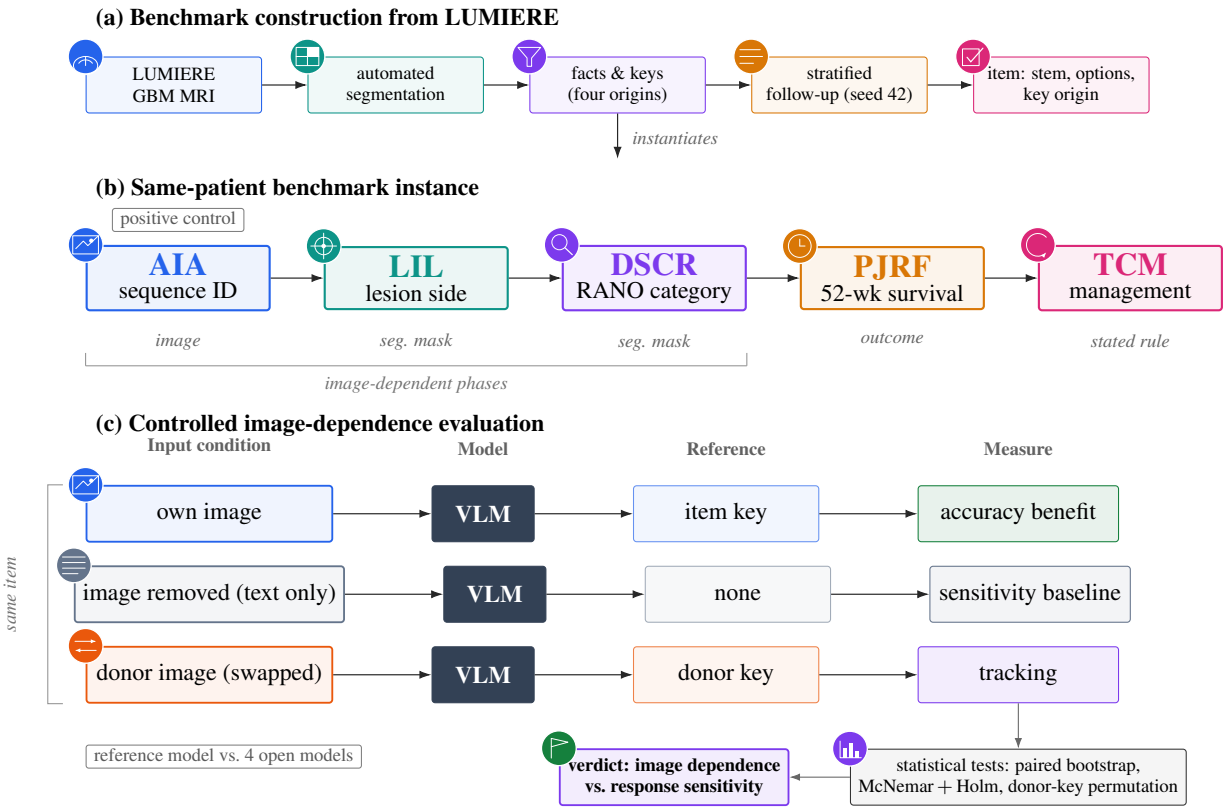


Figure 1: Evaluation overview. (a) Construction: a LUMIERE glioblastoma case becomes items through automated segmentation, fact extraction from the images, masks and records, stratified sampling of one follow-up, and item assembly; the keys have four origins (AIA sequence, LIL and DSCR segmentation, PJRF outcome, TCM rule). (b) The item instantiates a five-phase same-patient chain; the image-dependent phases (AIA, LIL, DSCR) carry the keys tested here and AIA is the positive control. (c) Each image-dependent phase is run under three controlled input conditions, the patient's own image, text only, and a donor-image swap, giving an accuracy benefit, a response-sensitivity baseline, and a donor-tracking measure; these feed the statistical tests and the image-dependence-versus-response-sensitivity verdict. A reference model is a positive control alongside the four open-weight models.

tion. This is an incompatibility between the released groupings and our evaluation, not a defect in a promise its authors made.

LUMIERE. LUMIERE (Suter et al., 2022) has 91 glioblastoma patients with longitudinal MRI (T1, contrast-enhanced T1, T2, FLAIR), automated segmentations (DeepBraTumIA and HD-GLIO-AUTO; Suter et al., 2022; Kickingereder et al., 2019), RANO ratings by one neuroradiologist (Wen et al., 2010), and overall survival for most patients. Segmentations are automated and were not manually verified; the ratings are expert-assigned.

3.1 Motivating audit: can an automated rule reproduce the expert ratings from the displayed slices?

The natural key for a diagnosis-category question is the expert RANO rating, but whether that rating can be recovered from the displayed slices is untested. We tested it with an automated reproducibility audit: for each first-line follow-up we tracked one enhancing lesion across scans, measured it on its own largest axial cross-section, computed the RANO bidimensional product and applied the enhancement criteria (progression: measurable disease at least 25% above the smallest earlier measurement, the nadir, or measurable disease where the nadir had none; response against the post-operative baseline; measurable means both diameters at least 10 mm). The rule has no new-lesion criterion beyond that. Coverage: of 241 first-line follow-ups, 44 are undetermined because no measurable enhancing disease exists on the baseline or nadir scan, so the rule abstains on them. Agreement, our primary number, is computed over the 197 determinate follow-ups: the rule reproduces the expert category for 75 (38%, Wilson 95% interval 32 to 45%,

and 29 to 48% when the 59 patients are resampled instead; counting the undetermined ones as failures would give 31%, which mixes abstention with disagreement). Most of the 122 disagreements are the rule returning complete response where the expert rated progressive or stable disease. Disagreement has several possible sources that this audit does not separate; LUMIERE records that one neuroradiologist rated the studies but not what they used. Our automated product correlates only moderately with the rater's recorded target-lesion diameters (Spearman 0.39 over 81 follow-ups), so measurement error is one plausible contributor, though the checkable subset was too small to establish a size-closeness difference either way (Appendix D). The audit therefore shows these ratings cannot be treated as validated targets for the displayed inputs; it does not measure how many are unanswerable, which would need qualified reviewers on a sample of the disagreements.

3.2 The item set

The keys have four different origins (Table 1), and we make no uniform claim that each is a function of what the model sees: AIA is the sequence rendered; LIL and DSCR are derived from automated segmentation masks the model does not see (recoverable from the slices only if the segmentation is right; these are segmentation-derived targets reviewers have not validated); PJRF is the recorded outcome, scored probabilistically since a future outcome is not a function of the stated facts; and TCM is the output of a stated management rule, testing adherence rather than clinical correctness. All random choices use fixed seeds and never look at model output.

Cohort. We select one first-line follow-up per patient (before any second surgery), in strata defined by the TCM rule class and the DSCR category, with quotas and seed fixed in advance, giving 46 patients (counts by phase and key in Appendix D). Selection used tabular records and segmentation masks only, and progressive-disease follow-ups within 2 weeks of the 12-week window are left out.

Images. Each image is an atlas-space axial slice, skull-stripped by the dataset authors with HD-BET (Isensee et al., 2019), at 4× magnification with a 20 mm scale bar, in radiological orientation. Slice levels are those on which the key is computed.

AIA, LIL, DSCR. AIA is the positive control: its key is the sequence rendered, balanced over four sequences (the most common covers 26.1% of items), so text alone can reach only chance in expectation. LIL's key is the hemisphere and anterior/posterior half of the largest tumor-core component on the displayed slice; lesions within 10 mm of a midline, smaller than 100 mm^2 or with under 80% of their pixels on their centroid's side of the midline are not asked about. DSCR's stem states the RANO enhancement criteria, and its key is the category computed from the bidimensional products.

PJRF. The prediction time is the follow-up scan and the outcome is death within 52 weeks of it, from the LUMIERE survival record. The model picks one of four probability bins (midpoints .10, .30, .50, .80) and is scored with the Brier score, not by whether it picked a "correct" bin. The base rate (Table 2) reflects our selection, not a population rate. The origin and censoring of the survival time are undocumented (Appendix F), so PJRF is secondary and no other claim in this paper rests on it.

TCM. LUMIERE records no treatment after chemoradiotherapy, so a documented treatment is not available and the key is a rule-based recommendation, not what was done: non-progressive disease continues adjuvant temozolomide; progressive disease within 12 weeks of completing radiotherapy is confirmed with repeat MRI before any change; later progressive disease prompts a change of therapy. The base stems supply the clinical facts but not this rule; only a post hoc condition states the rule in the prompt (Section 5). The 12-week window is from the RANO criteria (Wen et al., 2010); because we cannot assess the radiation field, the rule is a simplification (see Limitations). Radiotherapy dates are not in LUMIERE, so the stem states the weeks since chemoradiotherapy under an assumed standard schedule (ending 10 weeks after surgery). The same diagnosis therefore maps to different actions depending on the stated timing (7 repeat-MRI and 20 change-therapy items among 27 progressive-disease items), which lets a lookup of the category alone be told apart from use of the timing fact.

4 Controls and Pre-specified Analysis

The target is *correct, task-relevant image dependence under specified interventions*: whether a

Phase	Question	Model-visible inputs	Key	n
AIA	Which MRI sequence is shown?	one axial slice (T1, T1 after contrast, T2 or FLAIR)	the sequence rendered	46
LIL	Hemisphere and anterior/posterior half of the main tumor region	follow-up contrast-enhanced T1 slice at the level of the largest tumor region	position of the largest tumor-core component on that slice, relative to the midline and the brain's front-to-back midpoint	38
DSCR	RANO enhancement category at follow-up	baseline, follow-up and (when different from the baseline) nadir slices at the largest cross-section of the tracked enhancing lesion; the RANO thresholds are stated	category from the bidimensional products on those slices	46
PJRF	Probability of death within 52 weeks of this scan	age, sex, MGMT, IDH, extent of resection, weeks since surgery; alive at the scan	recorded outcome; scored by Brier score, not accuracy	45
TCM	Next management step	the same clinical facts plus weeks since chemoradiotherapy; the category comes from the chain	rule: non-PD continue; PD within 12 weeks of radiotherapy repeat MRI first; PD later change therapy	46

Table 1: The phases. Stems for AIA, LIL and DSCR contain no patient-specific fact and every option set contains every possible answer, so a swapped image never conflicts with the text and the donor's answer is always available. DSCR shows two images, or three when the nadir differs from the baseline.

model's answer moves toward what the displayed image supports, not merely whether its output reacts to the image. Majority-baseline performance is contextual evidence for an accuracy gain, not a necessary test for image use: a model can read some items from the slice yet still score below the most common key by missing others, so accuracy below that share neither establishes nor refutes image use on its own.

Models are the four open-weight models (MedGemma-4B (Sellergren et al., 2025), Gemma-3-12B and Gemma-3-27B (Gemma Team, 2025), Llama-4-Scout; 4-bit, greedy decoding, 800 tokens; the reference model uses 16,384 tokens because hidden reasoning counts against the cap; unparseable responses score incorrect). Each condition asks one question with a manufactured upstream context, so a phase's effect is not mixed with the model's own earlier answers.

- **E1, image effect** (AIA, LIL, DSCR): accuracy with the image minus text only, paired by patient, with a patient-bootstrap interval. The margin is ± 10 percentage points, fixed in advance; "within margin" requires the whole interval inside it, which at these sample sizes is reachable only when few items differ between the arms. Holm correction over the 12 model-phase tests; other contrasts are descriptive and unadjusted, with exact McNemar tests (McNemar, 1947) and Wilson intervals (Wilson, 1927) where a proportion is reported. Because the bootstrap collapses to zero width when every pair is concordant, we also report, post hoc, an add-half interval for the paired difference (Agresti and Min, 2005) and the exact p. **AIA is the positive control:** the plan's operational reading rule is that for a model whose unadjusted AIA interval is not above zero, the test has not been shown to detect image use, and E1 on LIL and DSCR supports no conclusion about it.

- **E2, donor tracking**: swap the image for a donor's with a different key (one donor per patient and phase, same number of images); $G = 1[\text{swap} = \text{donor key}] - 1[\text{text only} = \text{donor key}]$, so the no-image prior is subtracted. The estimand is conditional on the fixed assignment, in which a patient donates to at most two recipients. Because a positive gain can come from a generic response to any image rather than from the shown donor, we add a post hoc *conditional label-association* permutation: holding each recipient's observed answers fixed, it treats the assigned donor-key labels as exchangeable among recipients within each equal-image-count stratum, keeping a reused donor's recipients together, subject to the swap constraint that no recipient receives its own key, and asks whether answers match the assigned donor's key more often than a key drawn from a different donor. It does not reconstruct the image-assignment mechanism, so it licenses only that narrow comparison. On synthetic predictors with known behaviour, the permutation flags a genuine tracker but not a generic image-presence shift and holds its false-positive

rate at the nominal 5% under a constraint-valid no-tracking null (Table 6; the Type-I calibration of this permutation is in Appendix D).

- **E3, use of stated facts** (TCM): with the true context, move the stated timing across the 12-week window (*flip_window*) or replace the stated category by one that maps to another action class (*flip_label*). The registered summary was the share of answers equal to the rule's new answer; because that counts answers that had not changed, we instead report each item's transition relative to the true-context answer (Table 11) and take moved over informative items as the rate of interest. A category lookup alone reaches at most 84.8% accuracy (56.5% if it always confirms first) and cannot follow the timing flip.
- **E4, upstream context**: true, randomly wrong (a seeded incorrect option per earlier answer), or absent context.
- **E5, PJRF** (secondary; results in Appendix F): Brier score and skill against the leave-one-out base rate, and AUC.

We report three observable outcomes separately: accuracy benefit (E1), output sensitivity (the share of items whose answer letter differs between the image and text-only arms, and between the swapped and own image), and correct donor tracking (E2); none observes internal processing. Output sensitivity, the transition tables and the output-handling counts are post hoc, descriptive analyses. The ordinary five-phase chain (own answers as context) is run with and without images as a secondary analysis; each arm regenerates its upstream answers, so later-phase differences are total effects through the chain.

Reference and precision controls. Three of the four open models did not meet the operational AIA gate (Section 5), so we added three post hoc controls, each reported apart from the four-model Holm family and run under the same prompts, temperature 0 and decision rules (tags and digests in Appendix F). (i) A reference (positive-control) model, Gemini-3.6-Flash, on E1 and E2; empty API responses are counted and rerun (0 of 386 records), and, being an API model, it can change without notice. (ii) A precision control, Gemma-3-27B rerun in bf16; its reading rule, fixed in advance, attributed part of the gap to quantization only if the bf16 AIA interval rose clearly above the 4-bit model's, but the serving stack also differs from the Ollama runs, so precision and stack are confounded (we draw no conclusion). (iii) A non-Gemma control, Pixtral-12B (bf16), on E1 and E2 for AIA and LIL; its DSCR image conditions are excluded post hoc because the three-image prompt returned empty output for many records (Appendix F).

Phase	n	Chance	Most common key
AIA	46	25.0	26.1%
LIL	38	25.0	28.9%
DSCR	46	25.0	58.7%
TCM	46	25.0	43.5% (lookup $\leq$ 84.8%)
PJRF	45	(Brier)	base rate 62.2%

Table 2: Non-model references on the items: chance, share of the most common key, the ceiling of a category-only TCM lookup, and the PJRF base rate.

5 Results

5.1 Reference values without a model

Table 2 gives what constants and lookups reach. AIA is close to uniform over four answers, DSCR's most common category is progressive disease for 58.7% of items, and a category-only TCM lookup reaches at most 84.8%.

5.2 Image controls and stated facts

The reference model reads the image. Gemini-3.6-Flash identifies the sequence for 95.7% of patients with the image and 21.7% without (difference +73.9 points, 95% CI [+58.7, +87.0]), places the lesion for 100.0% against 36.8%, and with the image swapped gives the donor's answer for 95.7% (AIA) and 100.0% (LIL) of patients (Table 8). A model that reads the slices can therefore answer these items, so the tasks are feasible; this does not show that every item is valid, nor why a weaker model fails. DSCR is different: the reference model scores 71.7% with the image and 58.7% without, where the most common key covers 58.7% of items, so most of its text-only accuracy is the class prior and it recovers the key for only 71.7% of items. We therefore cannot establish that DSCR is answerable from the slices for every item (the shortfall may be perception, keys the model does not see, or key error), so a null DSCR effect for another model cannot be read as a failure to use the image.

Open models: image effect (E1). Sequence identification stays between 26.1 and 37.0% (chance 25%); Table 3 gives every cell and Figure 2 the per-model effect with its interval. The

Model	Phase	n	Image	Text-only	Δ [95% CI]	Add-half CI	Exact p	Answers differ	Interval
MedGemma-4B	AIA	46	34.8	26.1	+8.7 [−2.2, +21.7]	[−3.7, +20.4]	.289	30.4	○ inconclusive
MedGemma-4B	LIL	38	23.7	18.4	+5.3 [−5.3, +15.8]	[−5.8, +15.8]	.625	18.4	○ inconclusive
MedGemma-4B	DSCR	46	19.6	10.9	+8.7 [+2.2, +17.4]	[−0.5, +17.2]	.125	19.0	• CI above 0
Gemma-3-12B	AIA	46	26.1	26.1	+0.0 [−13.0, +13.0]	[−12.2, +12.2]	1.00	32.6	○ inconclusive
Gemma-3-12B	LIL	38	31.6	23.7	+7.9 [−7.9, +23.7]	[−9.3, +24.3]	.549	47.4	○ inconclusive
Gemma-3-12B	DSCR	46	58.7	47.8	+10.9 [+2.2, +19.6]	[+0.9, +20.0]	.062	11.9	• CI above 0
Gemma-3-27B	AIA	46	32.6	26.1	+6.5 [−2.2, +17.4]	[−3.6, +16.1]	.375	26.1	○ inconclusive
Gemma-3-27B	LIL	38	23.7	18.4	+5.3 [−5.3, +15.8]	[−5.8, +15.8]	.625	34.2	○ inconclusive
Gemma-3-27B	DSCR	46	58.7	58.7	+0.0 [+0.0, +0.0]	[−4.1, +4.1]	1.00	0.0	■ within ±10 pp
Llama-4-Scout	AIA	46	37.0	23.9	+13.0 [+2.2, +26.1]	[+0.8, +24.2]	.070	30.4	• CI above 0
Llama-4-Scout	LIL	38	39.5	34.2	+5.3 [−10.5, +21.1]	[−11.2, +21.2]	.754	44.7	○ inconclusive
Llama-4-Scout	DSCR	46	54.3	8.7	**+45.7 [+28.3, +63.0]**	[+26.0, +61.5]	<.001	88.1	• CI above 0

Table 3: E1: accuracy (%) with the image and text only (no upstream context), the paired difference in points with its bootstrap interval, an add-half sparse-data interval (Add-half CI), the exact McNemar p (unadjusted), the share of items whose answer differs between the two arms (answers differ), and the interval reading (unadjusted; only Llama-4-Scout on DSCR survives Holm correction). Answers-differ is over items in all three arms (AIA 46, LIL 38, DSCR 42 of 46); accuracy is over all items per phase (AIA 46, LIL 38, DSCR 46). AIA is the positive control.

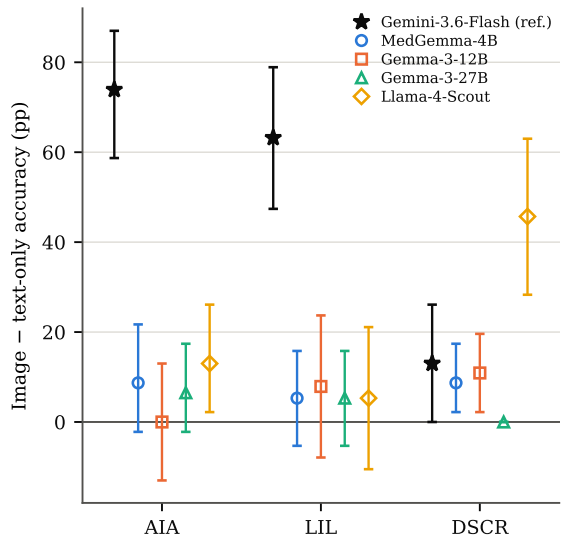


Figure 2: E1 image effect (image minus text-only accuracy, percentage points) per model by phase, with 95% bootstrap intervals; the same values as Table 3 (reference model from Table 8). The Gemini-3.6-Flash reference separates from the text-only arm on sequence and lesion side, whereas the four open models sit near zero, apart from Llama-4-Scout on the DSCR task, where a dominant class prior inflates the image-minus-text gap. Precision/stack controls are omitted for legibility and reported in the text.

operational AIA gate (an unadjusted interval above zero) is met only by Llama-4-Scout, and only on the bootstrap interval (exact McNemar $p = .070$). No conclusion depends on this borderline pass: the gate is a conservative screen, not a prerequisite for lesion or category reading, so the DSCR case below is read through the additions, and limited open-model image dependence holds whether or not Scout passes. One image-effect test survives Holm correction, Llama-4-Scout on DSCR, but it does not show the model reads the category from the slices: its accuracy with the image stays below the most common key's share (58.7%) and its text-only accuracy far below, so the difference gives the gap, not its mechanism. This below-majority accuracy is not a constant answer: class-wise (Table 5), Scout answers 19 of 27 progressive-disease items and recovers 6 of 19 non-majority items (complete-response recall 44%, stable-disease 33%), so scoring below the majority share reflects redistributed errors, not non-use of the image. The DSCR intervals above zero for MedGemma-4B and Gemma-3-12B do not survive correction; on LIL every interval includes zero. A test that fails correction is not evidence of no effect: no open-model AIA or LIL interval lies inside the ±10-point margin. Each open model's AIA answers concentrate on one sequence (modal answer 52 to 76% of the 46 items), consistent with a fixed default under weak image use, though every image reaches the model (Appendix F).

Precision control. Rerun in bf16, Gemma-3-27B identifies the sequence for 26.1% with the image and 26.1% without (against 32.6% at 4-bit; Table 8): the interval includes zero, so the rerun did not restore AIA performance, and its LIL and DSCR intervals also include zero. Because precision and serving stack changed together, we draw no conclusion about quantization. A different-family bf16/vLLM model, Pixtral-12B, is likewise near chance on sequence (30.4% against 26.1%) and lesion side, with no detected donor tracking (DSCR image arms excluded; Appendix F), so 4-bit Ollama serving is an unlikely sole cause, though scale and training still differ (full intervals and donor gains in Appendix E).

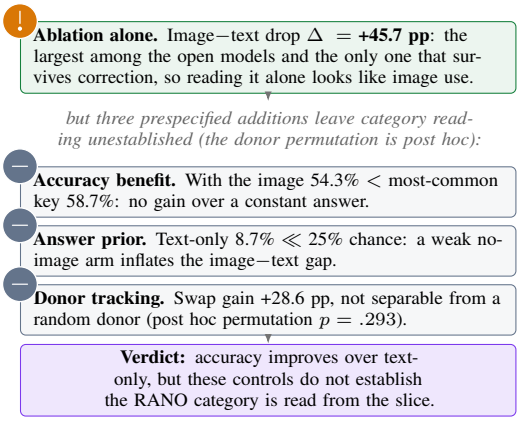


Figure 3: Worked example (Llama-4-Scout, DSCR): the ablation drop, largest among the open models and surviving correction, would alone credit image use, but the three prespecified additions (accuracy benefit over the answer prior, the prior itself, and donor tracking, the last a post hoc permutation) do not establish that the category is read from the slice.

Open models: donor tracking (E2). Swapping the image changes a substantial share of AIA and LIL answers (Table 7), but the new answer follows the donor's key with an interval above zero in only two cells: MedGemma-4B on AIA (gain +10.9; exact $p = .062$) and Llama-4-Scout on DSCR (+28.6; $p = .029$), the phase in which its accuracy gain also survives correction. Both survive resampling by donor and dropping any one donor. Llama-4-Scout on DSCR is not separable from a random assignment of donor keys (permutation $p = .293$); MedGemma-4B on AIA reaches a raw permutation $p = .040$ but does not survive Holm correction over the twelve open-model cells ($p = .482$) and fails the AIA positive-control gate, so we read neither as donor-specific tracking (Appendix D). Across 5 redrawn valid donor assignments Scout's DSCR gain ranges +19.0 to +33.3 points yet the permutation is significant in none of them (p from .057 to .548), so donor-specific tracking is not detected under any assignment, while the reference model stays significant throughout. Every other interval includes zero, and the open-model sequence and lesion-side gains are far below the reference model's (Table 8).

Worked example. Figure 3 works Llama-4-Scout on DSCR through the controls: the largest ablation drop among the open models, and the only one to survive correction, shows response sensitivity to the image, but the accuracy benefit, the answer prior and the post hoc donor permutation ($p = .293$, not separable from a random assignment) leave correct category reading unestablished.

Stated facts and context (E3, E4; extension). With the true context, TCM accuracy is below the best category-only lookup (84.8%) for all four open models, though Gemma-3-12B and Gemma-3-27B exceed the confirm-first lookup (56.5%). Reading each item against itself (Tables 10 and 11), replacing the stated category moves the answer for a minority of informative items in three models and a slim majority in Gemma-3-27B (25 of 43); crossing the 12-week window moves it most for Llama-4-Scout and not at all for Gemma-3-27B, and MedGemma-4B answers the most common class almost regardless. True context beats no context for three models and random wrong context for all four on TCM (unadjusted), but does not help on LIL or DSCR. Stating the rule in the prompt (post hoc) raises accuracy to 82.6% for Gemma-3-12B and 80.4% for Gemma-3-27B (from 69.6% unstated; paired gains +13.0 [+0.0, +26.1] and +10.9 [+2.2, +19.6]) and to 67.4% for Llama-4-Scout (+13.0), while MedGemma-4B does not move (+0.0); all stay at or below the best category lookup, so a supplied rule is followed incompletely even when stated (Appendix F).

Chain (secondary). The ordinary chain matters only for TCM, through the stated facts above; other image-minus-text differences are small with intervals including zero, so we claim no causal-degradation effect across phases.

6 Conclusion

Accuracy on an expert-labeled brain-MRI chain cannot show whether a model reads the scan when the labels are not validated targets for what is displayed: an automated rule reproduces the expert RANO rating for 38% of determinate follow-ups, and the source benchmark did not establish same-patient chains. On a LUMIERE chain with a stated key origin and fixed controls, a reference model reads sequence and lesion side, whereas four open-weight models show limited, inconsistent image dependence: one accuracy gain survives correction but stays below the most common key's share, no donor-specific tracking survives multiplicity correction, and none exceeds the best category lookup. These results do not identify internal image use; clinician review and a second reference model remain outstanding.

Limitations

Clinician review is outstanding. No clinician has verified that each question is answerable from the displayed inputs or that each key is correct. LIL and DSCR are derived from DeepBraTumIA segmentation masks unseen by the model, so segmentation errors propagate directly to these targets; the reference model supports task feasibility, not item-level validity. The enhancement rule further omits T2/FLAIR change, clinical status, and steroid dose, abstains when no measurable enhancing disease exists (44 of 241 follow-ups), and matches the expert category for 38% of determinate cases. Disagreement may reflect segmentation or measurement error, a different target lesion or reference scan, simplification of the rule, or evidence outside the displayed slices, which this audit cannot distinguish.

Our complete-response label is also weaker than clinical RANO: it is assigned whenever follow-up enhancement falls below the measurability threshold, potentially including residual sub-threshold disease, rather than requiring sustained disappearance of all enhancing and nonmeasurable disease. The rule tracks only the enhancing component nearest the baseline target lesion and does not implement RANO's new-lesion criterion, under which an unequivocal new lesion implies progressive disease; an audit of the follow-ups the rule labels complete response but the expert labels progressive disease (Appendix D) finds that this missing criterion accounts for some but not all of the disagreement. The expert rating is recorded for audit purposes but is not used as the benchmark key.

The management key is a rule, not a documented treatment. LUMIERE records no treatment after chemoradiotherapy. The TCM key therefore assumes a radiotherapy schedule ending 10 weeks after surgery and simplifies the 12-week RANO window, which can still permit progression when new enhancement lies outside the radiation field. PJRF further assumes that survival time and scan weeks share the surgery-week origin, while censoring is undocumented. With only 45 patients, a leave-one-out logistic model using the stated facts did not beat the base rate (Brier 0.283 vs. 0.246). This does not imply that the facts lack predictive information: the base rate reflects our stratified sample, and the outcome construction itself is unverified. We therefore treat PJRF as secondary and make no substantive claim from it.

Power and scope. With 38 to 46 patients per phase, the pre-specified ± 10-point margin is attainable only when few items differ between arms, so several comparisons remain inconclusive. Only one E1 image-effect test (Llama-4-Scout on DSCR) survives Holm correction. Moreover, MedGemma-4B, Gemma-3-12B, and Gemma-3-27B fail the preregistered AIA positive-control gate; under the analysis plan, their LIL and DSCR effects therefore support no conclusion. AIA tests only sequence identification, so passing it does not establish lesion reading. Donor images are from different patients and thus differ in anatomy beyond the manipulated feature, while text-only stems still refer to images.

The study uses a single-center, deliberately stratified cohort. The four open-weight models were served at 4-bit through Ollama, whereas the reference model is an API system of unknown size and precision, so between-model differences confound scale, training, quantization, and serving stack and cannot be attributed specifically to shortcut learning. bf16 reruns of Gemma-3-27B and a different-family model, Pixtral-12B, through vLLM also remained near chance on AIA, making 4-bit Ollama serving an unlikely sole explanation but not isolating quantization from serving, scale, or training; only these two models were rerun. The reference model was evaluated at temperature 0, once per condition on the original donor assignment (the swap condition was additionally rerun under alternative assignments; Appendix D), and API models may change over time.

Input handling was checked only in part. Post hoc checks confirmed that every image increased prompt token count, including the second and third DSCR images, and that each pipeline answered known-answer probes correctly (Appendix F). We did not verify that each serving stack passes the same image representation to its encoder as a reference implementation, and a single-patient check cannot establish preservation of MRI contrast across encoders. Thus, near-chance sequence identification in several open models remains unexplained: dropped images appear unlikely, but weak visual features or default-answer behavior remain possible. MedGemma-4B produced invalid JSON for 105 of 965 records and Llama-4-Scout for 46 of 965; pattern matching recovered the answer letter except for 0 MedGemma records (Appendix F). Intervals treat patients as exchangeable.

Ethics Statement

This work evaluates the diagnostic reasoning of general-purpose and medical-domain language models on real, de-identified patient imaging data; it does not deploy, recommend, or endorse any model for clinical use, and no results in this paper should be interpreted as a clinical validation of any evaluated system. LUMIERE is used under its published non-commercial research license, and all patient data is accessed in its already de-identified, publicly released form; no new identifiable patient information is collected. The LUMIERE authors report that the cantonal ethics committee of Bern approved the studies and waived written informed consent, that the images are skull-stripped, and that dates are given as weeks relative to the pre-operative scan; we did not attempt to re-identify patients. An AI coding assistant (Claude Code) helped write and run the analysis code and edit the text; the authors are responsible for all content.

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A OmniBrainBench Compatibility Details

A chain presupposes that a case's phases are one patient's trajectory. In the 6,823-question closed-ended subset of OmniBrainBench (Peng et al., 2026) only **2** verifiable cross-phase patient links exist (both in VQA_RAD, one AIA and one LIL question); stripping modality, view and slice suffixes from filenames gave the same result. The release has no complete patient identifier, so these are lower bounds. Each sub-corpus was built for one clinical task, so a patient is essentially asked one phase-type of question, and reading `source_file` as patient identity would stitch together different patients. This is an incompatibility with our evaluation, not a defect in the benchmark. The supplement's patient-level ADNI extension is not a brain-tumor cohort and, as of September 2026, has label-generation scripts only; we did not evaluate it. We found no dataset with same-patient, multi-phase continuity and a question layer (Section 2), which motivates LUMIERE.

B Artifacts, Licenses, Compute and Intended Use

Data. LUMIERE (Suter et al., 2022) (91 patients, from Figshare) may be used only for non-commercial purposes, which our research use respects; its authors report ethics approval and a consent waiver by the cantonal ethics committee of Bern, skull-stripped images and dates blurred to weeks. We share the rendered slices the models saw, with attribution and for non-commercial research only, not the imaging archive. OmniBrainBench (CC BY-SA 3.0 per its dataset card) was used only to count cross-phase links (Appendix A) and is not redistributed.

Reproducibility artifact. An anonymous repository (link in the submission) holds the items with keys and key origins, the donor assignments, the exact prompt of every single-question condition, every model response used, a manifest of which run supplied each record and which later records replaced earlier ones, the analysis code with fixed seeds, and one command that regenerates every statistic and table and checks them against this paper. It excludes model weights, API keys and the full LUMIERE archive.

Models and terms. MedGemma-4B (Health AI Developer Foundations terms), Gemma-3-12B and Gemma-3-27B (Gemma Terms of Use), Llama-4-Scout (Llama 4 Community License) and Gemini-3.6-Flash (Google's API terms) were used for research evaluation only, as these terms permit; no output is used clinically. Tags, digests and quantization are in Table 9. The MedGemma and Gemma 3 technical reports are cited only to identify those models; they are not peer-reviewed, and no claim rests on them. Llama-4-Scout has no technical report.

Software and compute. Ollama 0.33.0 served 4-bit Q4_K_M builds of the four open models (4.3,

12.2, 27.4 and 108.6 billion parameters; Llama-4-Scout is a mixture of experts with 17B active) and vLLM 0.27.1 the two bf16 runs (Gemma-3-27B and Pixtral-12B); the analysis used Python 3.12 with NumPy, SciPy, pandas and Pillow (exact versions pinned in the reproducibility artifact). Almost all GPU time was on one NVIDIA RTX PRO 6000 Blackwell GPU per job: about 154 GPU-hours for all LUMIERE work including failed and exploratory jobs, about 56 for the runs and input checks reported here. The Gemini runs made 525 API calls and about 1.1 million tokens including hidden reasoning tokens. Decoding was greedy with the caps in Section 4; no hyperparameters were tuned and each condition ran once, except that the swap condition was additionally rerun under 5 alternative donor assignments for the robustness check (Appendix D).

Intended use. The items, code and results are for evaluation research only: not a clinical benchmark, keys not clinician-validated, and nothing here supports using a model for diagnosis, prognosis or treatment.

C A Worked Example

The patient is Patient-007, the first in id order with all five phases and a three-image DSCR item (a rule fixed before any answer was inspected). Figure 4 shows the images the models see and the AIA and LIL donor images used in the swap (the DSCR donor slices are omitted for space); each item is four-way multiple choice, with the full stems and options in the reproducibility artifact.

AIA (positive control). Which MRI sequence does this image show? *Key:* D) T1-weighted, after gadolinium contrast.

LIL. Where is the main tumor region (enhancing tumor, necrosis or resection cavity, not the surrounding edema) on this slice? *Key:* A) Right hemisphere, posterior half.

DSCR. What is the response category at follow-up? *Key:* B) Progressive disease.

The key comes from the bidimensional products of the tracked lesion on the three slices (10, 0 and 385 mm^2 at baseline, nadir and follow-up); there is no measurable enhancing disease at the nadir and a measurable lesion at follow-up (a new measurable lesion), which the rule calls progressive disease. The expert rating is PD ("Target L.: 1, 16mm x 19mm"), the same category, so this is not an audit disagreement.

	AIA	LIL	DSCR
Key/donor	T1c/FLAIR	R post/L post	PD/PR
Gemini-3.6-Flash	T1c/T1c/FLAIR	R post/R ant/L post	PD/PD/CR
Llama-4-Scout	T1c/T1c/T1	L post/L post/L post	PD/SD/PD
Gemma-3-27B	T2/T1c/FLAIR	L ant/L ant/L post	PD/PD/PD
Gemma-3-12B	T2/T2/FLAIR	R ant/R ant/L post	PD/PD/PD
MedGemma-4B	T1/T1/T1	R ant/R ant/L ant	CR/CR/CR

Table 4: Answers for Patient-007 (own/text/swap in each cell), with the key and donor key; Gemini-3.6-Flash is the reference model. Labels: T1c is T1 after contrast; L/R hemisphere, ant/post half; PD, SD, PR, CR are RANO categories.

TCM. Using your earlier assessment of this patient, what is the most appropriate next step? *Key:* C) Change therapy now: bring re-resection, second-line treatment or a clinical trial to tumor board.

The base stem states the patient facts (see the artifact) but not the management rule; the rule is supplied in the prompt only in the post hoc explicit-rule condition (Section 5). The key follows that rule: non-PD $\rightarrow$ continue; PD and $\leq$ 12 weeks after radiotherapy $\rightarrow$ repeat MRI first; PD and later $\rightarrow$ change therapy (RANO 2010; Stupp schedule assumed). Here the category is progressive disease and chemoradiotherapy ended 53 weeks before the scan, so the class is escalate; changing the stated timing across the 12-week window or the stated category changes the rule's answer (the fact flips of E3).

PJRF (secondary). Same patient facts as TCM; the model forecasts the probability of death within 52 weeks of the scan (recorded outcome: death), scored by the Brier score.

Donor swap. Each image is replaced by a donor's whose key differs (AIA Patient-054, FLAIR not T1c; LIL Patient-023, L post not R post; DSCR Patient-019, PR not PD); the stem and options are unchanged, so the donor's answer is always an option.

D Item Set Summary and Donor-Dependence Checks

Item set. From 59 patients with a first-line follow-up we keep 46. AIA has 46 items (T1 12, T1 after contrast 12, T2 11, FLAIR 11). LIL has 38: the rest have no tumor core, a core within 10 mm of a midline or crossing it, or one under 100 mm^2 (posterior left 11, posterior right 11, anterior right 9,

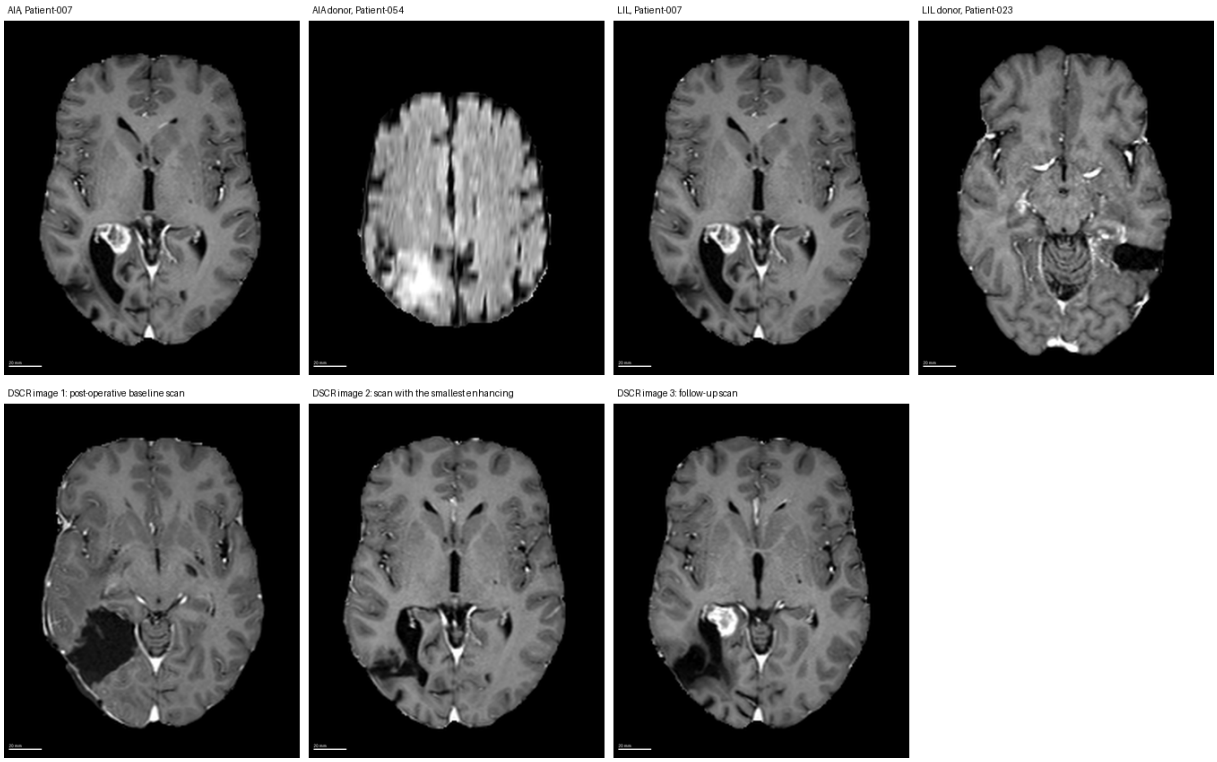


Figure 4: Images for Patient-007. Top: the AIA slice and its donor, the LIL slice and its donor. Bottom: the three DSCR slices (baseline, nadir, follow-up). Slices are atlas-space, skull-stripped, magnified 4×, with a 20 mm scale bar.

True key	PD	CR	SD	PR	Recall (%)
PD ($n = 27$)	19	4	2	2	70
CR ($n = 9$)	3	4	0	2	44
SD ($n = 6$)	4	0	2	0	33
PR ($n = 4$)	4	0	0	0	0

Table 5: DSCR class-wise confusion for Llama-4-Scout with its own image (rows: the item's RANO key class; columns: the model's predicted class; recall on the diagonal). Overall accuracy 54.3% is below the majority share (58.7%), but the model is not a constant predictor: it recovers 6 of 19 non-majority items while missing some progressive-disease items, so below-majority accuracy is not evidence that the slice is unread.

anterior left 7). DSCR has 46 (progressive disease 27, complete response 9, stable disease 6, partial response 4), the key equalling the expert rating for 26. TCM has 46 (continue 19, repeat MRI first 7, change therapy 20). PJRF has 45: a patient needs a recorded survival time after the scan week. A swap needs a donor with a different key and the same number of images; each patient gets one donor per phase and donates to at most two recipients, leaving 46 AIA, 38 LIL and 42 DSCR pairs (four patients have no compatible DSCR donor).

Audit size check. Where the rater recorded target-lesion diameters (81 follow-ups, 69 of them rated progressive disease), agreement was 21 of 33 where our product was within a factor of 1.5 of the rater's and 22 of 48 otherwise, a gap this subset is too small to establish as a real difference, so it neither rules in nor rules out size discrepancy as an explanation of the disagreements; it also has few stable-disease or partial-response ratings, where disagreement concentrates, so it cannot test them.

New-lesion disagreements (audit). The enhancement rule does not implement RANO's new-lesion criterion, under which an unequivocal new lesion implies progressive disease. Among the 46 follow-ups the rule labels complete response but the expert labels progressive disease, 22 contain a distinct newly measurable enhancing component elsewhere, 8 contain a similarly sized component already present at the nadir, and 16 contain no other measurable enhancing component, so the missing criterion accounts for some but not all of the disagreement.

Donor dependence (post hoc). AIA, LIL and DSCR use 31, 30 and 29 distinct donors respectively, none used more than 2 times; because donors

are reused across recipients, recipients that share a donor are not independent, so a naive component bootstrap is uninformative. We instead resample recipients that share a donor together, drop each donor in turn, and permute donor keys among recipients within each equal-image-count stratum, keeping each reused donor's recipients as a single block so that a donor shown to two recipients contributes its key to exactly two recipients under every permutation, and never onto a recipient's own key; p is the share with a gain at least as large. The permutation is a conditional label-association test and does not reconstruct the image-assignment mechanism. For DSCR it preserves the 24 two-slice and 18 three-slice items as separate strata; AIA and LIL have a single image-count stratum. MedGemma-4B on AIA: gain +10.9, donor-cluster interval [+2.1, +22.7], leave-one-donor-out +6.8 to +11.4, permutation $p = .040$. Llama-4-Scout on DSCR: gain +28.6, interval [+2.4, +53.7], leave-one-donor-out +25.0 to +35.0, permutation $p = .293$. One of the twelve open-model cells, MedGemma-4B on AIA, reaches a raw permutation $p = .040$; after Holm correction over the twelve cells it is not significant ($p = .482$), and MedGemma-4B does not pass the AIA positive-control gate, so no open-model cell shows donor-specific tracking.

Protocol validation with known-behaviour predictors. To illustrate that the donor-key permutation discriminates correct donor-specific tracking from a generic response to any image, we ran the same gain and permutation on synthetic predictors with known behaviour, on the real DSCR items and donor assignment (42 recipients; 50 simulations; detection rates reported with Wilson intervals against a nominal 5% level; no model is called). The permutation is the conditional label-association test of the donor-dependence check above, with the gain computed on the option texts so it does not depend on the lettering (options are already shuffled per item at load); it permutes the donor keys within each equal-image-count stratum and keeps a reused donor's recipients as one block, matching the structure of the real assignment, and remains a conditional label-association test that does not reconstruct the image-assignment mechanism. Its false-positive rate is calibrated by simulation (Section D).

Three no-tracking predictors that respond to image presence but not to which donor is shown serve as false-positive checks: a generic image shift (the most common donor class whenever an image is present) reaches a positive gain of +19.0 points yet is never flagged (0%), and predictors keyed to the option position or to the number of scans shown are flagged in no more than 7% of runs (upper Wilson bound), consistent with the nominal 5% level, though the simulation count leaves the interval too wide to establish a rate below it. A genuine tracker that returns the shown donor's key with probability ρ is flagged in 100% of runs at $\rho = 1$ and 78% at $\rho = 0.4$ (its effective probability of emitting the donor key is $\rho + (1 - \rho)/4 = 0.55$, since the guess branch can also land on it). The same separation holds on AIA and LIL (the generic shift never flagged; the noisy tracker flagged in 96% and 96% of runs) and across re-drawn valid assignments (Table 6 gives DSCR). The gain alone does not separate tracking from a generic image response, since the generic shift also has a positive gain; the permutation does, flagging the tracker but not the shift. These checks illustrate the test's discrimination; a non-significant result for the open models indicates that we did not detect donor-specific tracking and does not rule out weaker or differently structured tracking.

Synthetic predictor	Donor gain (pp)	Answer change (%)	Detected ($p < .05$)
Constant answer	+0.0	0	0%
Random answer	+2.4	79	2%
Generic image shift	+19.0	100	0%
Scan-position bias	−2.4	64	0%
Scan-count bias	−9.5	57	0%
Tracker ($\rho = 1$)	+66.7	45	100%
Tracker ($\rho = 0.7$)	+45.2	55	100%
Tracker ($\rho = 0.4$)	+19.0	63	78%

Table 6: Diagnostics on the DSCR donor swap (42 recipients), run on synthetic predictors with known behaviour over 50 simulations: the donor-tracking gain, the image-vs-text answer-change rate, and the share of simulations in which the permutation flags tracking ($p < .05$). A generic image shift that ignores which donor is shown reaches a positive gain and a high answer-change rate yet is never flagged, and at $\rho = 0.4$ a genuine tracker and the generic shift share the same gain (+19.0 vs +19.0 pp): neither the gain nor the answer-change rate separates them, the permutation does.

Validity of the stratified donor permutation (post hoc). The permutation respects the structure of the real donor assignment: it shuffles the donor keys only within an equal-image-count stratum (for DSCR, the 24 two-slice and 18 three-slice items), keeps each reused donor's recipients as one block, and never assigns a recipient its own key. We check that this samples the intended conditional null (no model is called). The null hypothesis is

that a recipient's answer is independent of which donor's key it was assigned; we hold the observed swap and text-only answers fixed. Drawing a fresh constraint-valid assignment for the real answers as a no-tracking truth and testing with this permutation, the test shows no excess false positives over 1,000 null datasets (rejection at or below the nominal 5%): 3.9% [2.9, 5.3] on DSCR (Llama-4-Scout) and 3.8% [2.8, 5.2] on AIA (MedGemma-4B), with median null p .567 and .611. We do not claim validity beyond the no-tracking mechanisms simulated here.

Robustness to the donor assignment (post hoc). Many assignments satisfy the swap constraints, so we re-ran the models under 5 further distinct valid assignments, drawn by the same procedure, and recomputed the gain and permutation. The reference model stays strongly significant on both phases (permutation $p < .001$ in 5 of 5 AIA and 5 of 5 LIL assignments; gain +65.2 to +84.8 on AIA and +65.8 to +86.8 on LIL). Llama-4-Scout on DSCR, the open-model cell re-run here, is significant in none of the 5 assignments (gain +19.0 to +33.3, permutation p from .057 to .548), so donor-specific tracking is not detected for this cell under any redrawn assignment.

E Donor-Swap and Reference-Model Tables

Tables 7 and 8 give the donor-swap results for the open models and the reference-model results.

Precision-control detail. The two bf16/vLLM models' image effect and donor gain (percentage points; bootstrap interval on the effect): Gemma-3-27B-bf16 is +0.0 [−10.9, +10.9] on AIA with donor gain +0.0, and +0.0 on LIL; Pixtral-12B is +4.3 [−4.3, +13.0] on AIA with donor gain −2.2, and −2.6 on LIL (its DSCR image arms are excluded; Appendix F). Every effect interval includes zero.

F Output Handling, Input Check and Fact-Flip Transitions

These are post hoc, descriptive checks. Invalid JSON has its answer letter recovered by pattern (the prompt orders it first). Of 965 records per open model (386 for the API and bf16 controls), only MedGemma-4B (105, truncation at the 800-token cap after a repeated sentence), Llama-4-Scout (46, a misspelled field name in otherwise complete output) and 1 Gemma-3-27B case produced invalid JSON; every letter was still recovered (0 and 0 unrecoverable), and Gemma-3-12B and the controls had none. The one exception is the second precision/stack control, Pixtral-12B: its DSCR image conditions returned empty output with no recoverable letter for 22 of 46 own-image and 18 of 42 swapped-image records and are excluded (its AIA and LIL arms, at most two empty per arm, are reported). MedGemma-4B's DSCR accuracy on valid-JSON records (23.1%) stays below nominal chance, close to its all-records score (19.6%), so parsing does not explain its below-chance score, which comes from the letters it chooses (never the fourth option). These counts do not show that images reach each model; the input check does.

Input check. For each open-model pipeline and the bf16 Gemma-3-27B rerun (the pipelines in Table 9) we sent one real AIA, one LIL and one three-image DSCR item with zero to three of its images behind a unique nonce (preventing prefix caching) and read the prompt token count. Each added image raises it by a similar amount (Table 9), so none was dropped, including the second and third DSCR images. These same pipelines passed known-answer probes (solid colours, printed numbers, image order and count, a slice's black background), and one slice stayed legible squashed to 896 and 448 pixels. The Pixtral-12B and API-reference pipelines were not token-probed: Pixtral's unusable DSCR image arms are excluded on output grounds above, and the API model's internal handling cannot be inspected this way. Run once on one patient, this shows images reach the model and simple stimuli are perceived, not that MRI contrast survives each image encoder.

Table 11 gives each item's transition under a fact flip relative to the same item under the true context: moved to the rule's new answer, already there, unchanged elsewhere, or changed to another answer. Only the first column shows movement to the new target. The last column is the registered summary: the share of all items whose flipped-context answer is the rule's new answer, counting items that moved to it together with items already there that stayed, but not items that began at the new answer and then moved away under the flip. It is highest for MedGemma-4B on the label flip, whose 21 already-at-target answers all stayed; for Gemma-3-27B and Llama-4-Scout, 3 and 3 already-at-target

Model	Phase	n	Own	Swap: donor	Text: donor	Gain [95% CI]	Add-half CI	Exact p	Swap changes
MedGemma-4B	AIA	46	34.8	32.6	21.7	+10.9 [+2.2, +19.6]	[+0.9, +20.0]	.062	32.6
MedGemma-4B	LIL	38	23.7	21.1	28.9	−7.9 [−18.4, +2.6]	[−19.3, +4.3]	.375	13.2
MedGemma-4B	DSCR	42	21.4	28.6	23.8	+4.8 [−4.8, +14.3]	[−5.3, +14.4]	.625	9.5
Gemma-3-12B	AIA	46	26.1	17.4	17.4	+0.0 [−13.0, +13.0]	[−12.2, +12.2]	1.00	50.0
Gemma-3-12B	LIL	38	31.6	23.7	18.4	+5.3 [−10.5, +18.4]	[−9.6, +19.6]	.727	52.6
Gemma-3-12B	DSCR	42	54.8	35.7	31.0	+4.8 [−4.8, +14.3]	[−5.3, +14.4]	.625	2.4
Gemma-3-27B	AIA	46	32.6	34.8	30.4	+4.3 [−4.3, +13.0]	[−4.9, +13.2]	.625	32.6
Gemma-3-27B	LIL	38	23.7	26.3	21.1	+5.3 [−7.9, +18.4]	[−7.9, +17.9]	.688	44.7
Gemma-3-27B	DSCR	42	54.8	35.7	33.3	+2.4 [−4.8, +9.5]	[−6.6, +11.2]	1.00	7.1
Llama-4-Scout	AIA	46	37.0	30.4	28.3	+2.2 [−6.5, +13.0]	[−7.9, +12.1]	1.00	32.6
Llama-4-Scout	LIL	38	39.5	31.6	23.7	+7.9 [−7.9, +23.7]	[−7.8, +22.8]	.508	55.3
Llama-4-Scout	DSCR	42	54.8	50.0	21.4	+28.6 [+7.1, +50.0]	[+5.6, +49.0]	.029	47.6

Table 7: E2: share of answers (%) that equal the donor's key when the image is swapped, and when there is no image; the gain is the difference in percentage points, with its bootstrap interval, an add-half sparse-data interval (Add-half CI) and the exact McNemar p (unadjusted); the last column is the share of items whose answer differs from the own-image answer after the swap. Sample sizes are the donor-paired items (AIA 46, LIL 38, DSCR 42); the own-image accuracy in Table 3 is over all items (DSCR 46).

Phase	n	Image	Text-only	Δ [95% CI]	Swap: donor key	Gain [95% CI]
AIA	46	95.7	21.7	+73.9 [+58.7, +87.0]	95.7	+67.4 [+54.3, +80.4]
LIL	38	100.0	36.8	+63.2 [+47.4, +78.9]	100.0	+86.8 [+76.3, +97.4]
DSCR	46	71.7	58.7	+13.0 [+0.0, +26.1]	50.0	+16.7 [+2.4, +31.0]

Table 8: Reference model (Gemini-3.6-Flash): accuracy (%) with the image and with text only, the paired difference, the share of swapped-image answers equal to the donor's key, and the donor-tracking gain (percentage points). Accuracy is over all items (AIA 46, LIL 38, DSCR 46); the swap and donor-key columns use the donor-paired items (AIA 46, LIL 38, DSCR 42).

items moved away under the flip, so their registered share falls below the moved-plus-already-there total.

PJRF (E5, secondary). The outcome's time origin and censoring are unverified: we assume the survival time shares the week numbering's origin, LUMIERE states neither, 4 of 86 patients have a survival time shorter than their last imaging week (1 in our cohort), and a four-week shift of the origin would flip 4 of the 45 labels. No model's Brier score differs clearly from the constant base rate (paired difference to the leave-one-out base rate, negative better, from −0.017 to +0.009 across the four models), and a leave-one-out logistic model on age, MGMT, resection extent and weeks since surgery does not beat it either (Brier 0.283 against 0.246, AUC 0.387); MedGemma-4B and Gemma-3-12B place 78% and 91% of patients in the 0.5 bin, so their Brier is near 0.25 by construction. We draw no conclusion from PJRF (Section 3.2); this small-sample null does not show the facts carry no information.

Model	Tag	Digest	Quantization	Tokens per image	Probes passed
MedGemma-4B	medgemma:4b	9fe4e9a6c9bd	Q4_K_M	285–308	8/8
Gemma-3-12B	gemma3:12b	f4031aab637d	Q4_K_M	284–305	8/8
Gemma-3-27B	gemma3:27b	a418f5838eaf	Q4_K_M	280–304	8/8
Llama-4-Scout	llama4:scout	bf31604e25c2	Q4_K_M	1456–1477	8/8
Gemma-3-27B (bf16)	google/gemma-3-27b-it	–	bf16 (vLLM)	271–288	8/8

Table 9: Input check per pipeline: model tag, digest and quantization as reported by the server, the range of prompt-token increments per added image, and known-answer probes passed out of eight.

	TCM acc. (%)			Moved / informative	
	true	wrong	none	label (all)	timing (PD)
MedGemma-4B	54.3	39.1	34.8	10/25	1/27
Gemma-3-12B	69.6	30.4	39.1	18/40	8/21
Gemma-3-27B	69.6	32.6	45.7	25/43	0/20
Llama-4-Scout	54.3	37.0	45.7	10/34	11/21

Table 10: E3: TCM accuracy (%) under true, random wrong and absent upstream context, and the number of answers that move to the rule's new answer after the stated category is replaced (label, all items) or the stated timing crosses the 12-week window (timing, progressive-disease items), over the informative items, those whose true-context answer was not already the new answer.

Model	Flip	Items	Moved	Already there	Unchanged	Other	Moved / informative	Registered share (%)
MedGemma-4B	label (all)	46	10	21	12	3	10/25	67.4
MedGemma-4B	timing (PD)	27	1	0	23	3	1/27	3.7
Gemma-3-12B	label (all)	46	18	6	10	12	18/40	52.2
Gemma-3-12B	timing (PD)	27	8	6	11	2	8/21	51.9
Gemma-3-27B	label (all)	46	25	3	2	16	25/43	54.3
Gemma-3-27B	timing (PD)	27	0	7	20	0	0/20	25.9
Llama-4-Scout	label (all)	46	10	12	15	9	10/34	41.3
Llama-4-Scout	timing (PD)	27	11	6	10	0	11/21	63.0

Table 11: TCM fact flips: transitions relative to the true-context answer for the same item. The label flip covers all 46 items; the timing flip covers the 27 progressive-disease items.